

A brief overview of the architecture

Most of the current DCT systems are designed as smartphone apps that communicate with one or more back-end servers. So, in most countries in the world, Bob and his close contacts must have smartphones to take advantage of any automated contact tracing system. People who carry feature phones or do not have personal mobile devices (including many elderly individuals or children), would still have to rely upon the manual contact tracing process. DCT apps are also referred to as Covid-19 apps.

The core functionality of an automated contact tracing system is to reliably detect and notify close encounters with infected individuals without revealing anyone's identity in the process. Since identity cannot be revealed, the apps are designed not to share Personally Identifiable Information (PII) like name, location, phone number, or device identification number (such as IMEI) between any two devices.

During close encounters, each app shares pseudo-random byte-strings (also called Proximity Identifiers, or Ephemeral IDs) with other apps and logs the byte-strings that it receives from the nearby apps in its local device storage. These pseudo-random numbers act as pseudonymous identifiers (and not as anonymous identifiers as they can be mapped back to the originating secret keys) and are rotated at a regular frequency so that the movement of devices cannot be tracked through any of these identifiers. Fig. 1 illustrates a sample exchange of 16-byte pseudonymous identifiers between two devices (A and B) having the same Covid-19 app installed in each of them.

Conceptually, a Covid-19 app can be deemed to have three layers, where the topmost layer (say Layer-1) takes care of the User Interface (UI), the middle layer (say Layer-2) is called the API layer (most of the encoding-decoding and encrypting-decrypting functionalities reside in this layer), and the bottom-most layer (Layer-3) is the link layer that supports the packet communication.

Layer-3 has two parts, each of

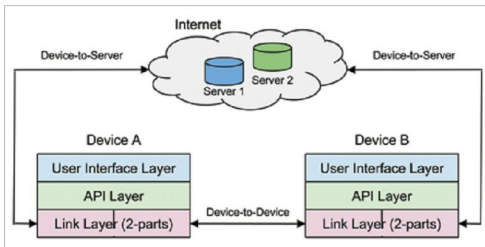


Fig. 2: Basic architecture

which takes care of a particular type of communication. One type of information exchange occurs between the mobile device and the back-end server(s) through the encrypted mobile data network. The other type of packet exchange takes place between the neighbouring devices that have the DCT app installed. In this article, we primarily focus on the technology and protocol used in this latter type of communication. Fig. 2 illustrates a basic schematic for this architecture.

There are three crucial requirements for device-to-device communication in any DCT system. First, it should take place when the devices are physically near each other so that a device can be realistically considered a proxy to the person carrying it for the purpose of detecting close contact events. Second, no device should be able to discover the identity of any other device, or for that matter any personally identifiable information of its user. Third, the communication should take place autonomously, without an active intervention of the user at every step. Also, the overhead of such information exchange should be low.

When we consider the above requirements, the usual channels of communication like mobile-to-mobile call, SMS, or any other instant messaging service would have to be ruled out. What technology can then be used to meet the needs of DCT apps? As it turns out, some of the possible candidates are Wi-Fi, ultrasound, and Bluetooth.

A group of researchers has created

prototypes, deployed the same in a couple of university campuses, and published their results in a paper in the month of May 2020. In this document they have proposed "a network-centric approach for contact tracing that relies on passive Wi-Fi sensing." Also, a company named Blynscy claims to have come up with a contact tracing technology using Wi-Fi.

Another approach is to use ultrasound, as it has been observed that smartphones emit ultrasound signals. Currently, NOVID is the only team in the world that has come up with an app that utilises ultrasound and provides sub-metre level accuracy in contact detection. However, the NOVID app has Wi-Fi and Bluetooth capabilities as well.

BLE in DCT systems

While there are examples of Covid-19 apps that utilise Wi-Fi and ultrasound technologies, a large majority of these systems use Bluetooth as the underlying technology of communication. Within Bluetooth also, there is a low energy variant, called Bluetooth Low Energy (BLE), that has caught the attention of most of the system designers and implementers.

Bluetooth Special Interest Group (Bluetooth SIG) originally proposed BLE as a wireless personal area network technology for "novel applications in the healthcare, fitness, beacons, security, and home entertainment industries." BLE is natively supported by a host of mobile operating systems including iOS, Android, and Windows Phone. It consumes much less power than the classic Bluetooth,